

## SAFETY DATA SHEET

Creation Date 10-Aug-2009

Revision Date 29-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Potassium hexacyanoferrate(III)  
**Cat No. :** 33357  
**CAS No** 13746-66-2  
**Synonyms** Red prussiate; Potassium iron(III)cyanide; Potassium hexacyanoferrate (III)  
**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

Classification under 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

#### Label Elements

##### **Hazard Statements**

##### **Precautionary Statements**

##### **Hazards not otherwise classified (HNOC)**

Contact with acids liberates very toxic gas

### 3. Composition/Information on Ingredients

| Component              | CAS No     | Weight % |
|------------------------|------------|----------|
| Potassium ferricyanide | 13746-66-2 | >95      |

### 4. First-aid measures

|                                            |                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                              |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Get medical attention.                                                                          |
| <b>Most important symptoms and effects</b> | No information available.                                                                                               |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                   |

### 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

#### Hazardous Combustion Products

Potassium oxides. Metal oxides. Hydrogen cyanide (hydrocyanic acid).

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
1

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

### 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing.                                                                                                             |
| <b>Environmental Precautions</b> | Avoid release to the environment. See Section 12 for additional Ecological Information. Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system. |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

|                 |                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from direct sunlight. Incompatible Materials. Strong oxidizing agents. Strong acids.             |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component              | ACGIH TLV                | OSHA PEL                                                                 | NIOSH                                                  | Mexico OEL (TWA)         |
|------------------------|--------------------------|--------------------------------------------------------------------------|--------------------------------------------------------|--------------------------|
| Potassium ferricyanide | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup> | IDLH: 25 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

|                             |                                                                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Measures</b> | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

### Personal Protective Equipment

|                                 |                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face Protection</b>      | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                   |
| <b>Respiratory Protection</b>   | No protective equipment is needed under normal use conditions.                                                                                                              |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                      |

## 9. Physical and chemical properties

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Physical State</b>                         | Crystalline Solid        |
| <b>Appearance</b>                             | Orange - Red             |
| <b>Odor</b>                                   | Odorless                 |
| <b>Odor Threshold</b>                         | No information available |
| <b>pH</b>                                     | ~ 6 5% aq. sol           |
| <b>Melting Point/Range</b>                    | No data available        |
| <b>Boiling Point/Range</b>                    | No information available |
| <b>Flash Point</b>                            | No information available |
| <b>Evaporation Rate</b>                       | Not applicable           |
| <b>Flammability (solid,gas)</b>               | No information available |
| <b>Flammability or explosive limits</b>       |                          |
| Upper                                         | No data available        |
| Lower                                         | No data available        |
| <b>Vapor Pressure</b>                         | negligible               |
| <b>Vapor Density</b>                          | Not applicable           |
| <b>Density</b>                                | 1.86 g/cm <sup>3</sup>   |
| <b>Specific Gravity</b>                       | No information available |
| <b>Bulk Density</b>                           | 1.05 kg/m <sup>3</sup>   |
| <b>Solubility</b>                             | Partially soluble        |
| <b>Partition coefficient; n-octanol/water</b> | No data available        |
| <b>Autoignition Temperature</b>               | No information available |

|                           |                                                 |
|---------------------------|-------------------------------------------------|
| Decomposition Temperature | > 200°C                                         |
| Viscosity                 | Not applicable                                  |
| Molecular Formula         | C <sub>6</sub> Fe K <sub>3</sub> N <sub>6</sub> |
| Molecular Weight          | 329.26                                          |

## 10. Stability and reactivity

|                                  |                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                        |
| Stability                        | Stable under normal conditions. Sensitivity to light.                             |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Excess heat. Exposure to light.      |
| Incompatible Materials           | Strong oxidizing agents, Strong acids                                             |
| Hazardous Decomposition Products | Potassium oxides, Metal oxides, Hydrogen cyanide (hydrocyanic acid)               |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                          |
| Hazardous Reactions              | Contact with acids liberates very toxic gas. Heating can release hazardous gases. |

## 11. Toxicological information

### Acute Toxicity

**Product Information** If ingested: the ferricyanide complex does not decompose to cyanide.

#### Component Information

| Component              | LD50 Oral                  | LD50 Dermal | LC50 Inhalation |
|------------------------|----------------------------|-------------|-----------------|
| Potassium ferricyanide | LD50 = 2,970 mg/kg (Mouse) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component              | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------|------------|------------|------------|------------|------------|------------|
| Potassium ferricyanide | 13746-66-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known  
**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

### Endocrine Disruptor Information

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not empty into drains. Do not allow material to contaminate ground water system.

| Component              | Freshwater Algae | Freshwater Fish                             | Microtox   | Water Flea                           |
|------------------------|------------------|---------------------------------------------|------------|--------------------------------------|
| Potassium ferricyanide | Not listed       | Onchorchynchus mykiss:<br>LC50: 869 mg/L/96 | Not listed | Daphnia magna: EC50: 549<br>mg/L/48h |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <u>DOT</u>      | Not regulated |
| <u>TDG</u>      | Not regulated |
| <u>IATA</u>     | Not regulated |
| <u>IMDG/IMO</u> | Not regulated |

## 15. Regulatory information

### United States of America Inventory

| Component              | CAS No     | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|------------------------|------------|------|--------------------------------------------------|--------------------------------|
| Potassium ferricyanide | 13746-66-2 | X    | ACTIVE                                           | -                              |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)** Not applicable

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component              | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------------------|------------|-----|------|-----------|-------|------|------|------|-------|----------|
| Potassium ferricyanide | 13746-66-2 | X   | -    | 237-323-3 | X     | X    | X    | X    | X     | KE-34764 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component | CAS No | Weight % | SARA 313 - Threshold | SARA 313 - Reporting |
|-----------|--------|----------|----------------------|----------------------|
|-----------|--------|----------|----------------------|----------------------|

|                        |            |     | Values % | thresholds |
|------------------------|------------|-----|----------|------------|
| Potassium ferricyanide | 13746-66-2 | >95 | 1.0 %    | -          |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component              | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|------------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Potassium ferricyanide | -                          | -                           | X                      | X                         |

**Clean Air Act**

| Component              | HAPS Data | Class 1 Ozone Depleters | Class 2 Ozone Depleters |
|------------------------|-----------|-------------------------|-------------------------|
| Potassium ferricyanide | X         |                         | -                       |

**OSHA** - Occupational Safety and Health Administration Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component              | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|------------------------|---------------|------------|--------------|----------|--------------|
| Potassium ferricyanide | -             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
 DOT Marine Pollutant N Y  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component              | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Potassium ferricyanide | 13746-66-2 | -                                                                   | -                                                                             | -                                                                                                     |

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component | CAS No | OECD HPV | Persistent Organic | Ozone Depletion | Restriction of |
|-----------|--------|----------|--------------------|-----------------|----------------|
|-----------|--------|----------|--------------------|-----------------|----------------|

|                        |            |                | Pollutant      | Potential      | Hazardous Substances (RoHS) |
|------------------------|------------|----------------|----------------|----------------|-----------------------------|
| Potassium ferricyanide | 13746-66-2 | Not applicable | Not applicable | Not applicable | Not applicable              |

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

#### Other International Regulations

| Component              | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Potassium ferricyanide | 13746-66-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y33                      |

## 16. Other information

#### Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

#### Creation Date

10-Aug-2009

#### Revision Date

29-Mar-2024

#### Print Date

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#### Revision Summary

New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**